Returns and Volatility Spillovers in the Currency Futures Markets: The Case of the German Mark and the Japanese Yen

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Abstract

The study provides evidence of the nature of the volatility transmission for daily currency futures contracts traded at the International Monetary Market (IMM) and the Singapore Exchange (SIMEX). The samples of the German mark currency futures contracts and the Japanese yen currency futures contracts are used to study inter-market spillover effects in the non-overlapping time zones. Employing GARCH specifications, the results find strong volatility transmission from the IMM to the SIMEX for both currency futures contracts but failed to find the reverse. This result suggests that currency futures market with larger trading volume i.e. IMM are influencing the direction of transmission of smaller currency futures market i.e. SIMEX

JEL classification: G13, G14

Keywords: Currency futures, return, volatility, spillover

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1. 1 Introduction

The increased globalisation in financial markets brought about by the relatively free flow of goods and capital as well as the revolution in information technology in recent years resulted in voluminous flow of information from one market to another. A number of studies have discussed the theoretical issue and documented empirical evidence related to the volatility contagion effect separated by distance as well as time. For example, Hamao, Masulis and Ng (1990) study on the international stock markets using Autoregressive Conditional Heteroscedasticity (ARCH) family of statistical models on daily opening and closing prices. The same idea is explored by Eun and Shim (1989), King and Wadhwani (1990), Cheung and Kwan (1991), Susmel and Engle (1994), and Liu, Pan and Fung (1996). They find evidence of volatility spillover from New York, *i.e.* US market to the other markets but not in opposite directions However, Lin, Engle and Ito's (1994) study on interaction of the US and the Japanese stock market finds that price and volatility spillovers are generally reciprocal *i.e*. the two markets influence each other.

As for the foreign exchange market, Engle, Ito and Lin (1990) have developed the hypotheses of heat waves and meteor shower and tested them on the daily exchange rate in the New York and Japanese markets. Their findings support the meteor shower, suggesting spillovers from one marker to another market. Similar finding also reported in Ito, Engle and Lin (1992).

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Studies on transmission for futures market are relatively few. Abhayankar (1995), for example, investigates the Eurodollar futures markets spillover and finds that there exists lagged spillover effect in the mean return from the CME to SIMEX but not vice-versa. Similar evidence is also found by Lim, Terry and How (1998) who examine volatility transmission between these market but using Eurodollar futures interest rate changes.

In currency futures markets, Najang, Rahman and Yung's (1992) study was the first to examine the volatility spillover employing Generalized Autoregressive Conditional Heteroscedasticity (GARCH) methodology. Specifically, they explore the extent to which foreign currency futures shocks impinge upon other currencies and the channels through which they are transmitted. Their sample consisted of four currency futures over 10 years from January 1980 to December 1989. The data set are from Chicago Mercantile Exchange contains daily data of the British pound, the German mark, the Japanese yen and the Swiss franc. In general, their findings implied that all currencies are involved at some periods in transmitting volatility to other currencies.

This paper uses the daily currency futures contracts of German mark and Japanese yen to examine the nature of the volatility transmission across the International Monetary Market of Chicago Mercantile Exchange (CME) and the Singapore Monetary Exchange (SIMEX). As argued by Lim, Terry and How (1998), since the traded contract at the IMM and at the SIMEX are identical by virtue of the mutual offset agreement and these two market do not overlap, the information about contract in one market is highly relevant and has an impact for the same contract in other market, resulting in volatility transmission. Furthermore, the study conducts an unbiased test of impact of news on the

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same assets but to different markets of different size, unlike most of the previous studies on the volatility spillover in currency futures which employ different currency futures but traded in the same exchange market (see Najang, Rahman and Yung, 1992).

Research in this area is importance to the market participants since it provides useful information about the market behaviour in difference exchange location with a difference time zone. This is true with the advancement of electronic trading systems, which provides the local investors to trade currency futures not only in local exchange but also from exchange, which is located elsewhere and outside the normal trading hour. For instance, this paper focuses on the trading arrangement between the CME and the SIMEX which allows the transfer of a market position executed in one exchange to another exchange and vice versa. Beside that, the investors can hold portfolios of securities in many exchanges and trade them whenever possible using difference exchanges markets. Furthermore they can monitor the dynamic of international price movements in difference exchange markets and thus, enable them to make quick buy/sell decision.

The remainder of this paper proceeds as follows; Section 2 describes our data and methodology. Section 3 discusses the empirical results. Summary and conclusions are given in Section 4.

1.2 Data and Methodology

Description of Data Set

The currency futures contracts were first traded at the International Monetary Market (IMM) of Chicago Mercantile Exchange (CME) in 1972. They are the first financial

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futures contracts to be introduced. Seven currencies are traded in this market: Australian dollar, British pound, Canadian dollar, German mark, French franc, Japanese yen and Swiss franc. The delivery months for the IMM currency futures contracts are September, December, March and June.

The Singapore Mercantile Exchange (SIMEX), which is a partner of the CME introduced an identical contract in September 1984 but limited only to two currency futures contracts of German mark and Japanese yen. Since the SIMEX is relatively new, their average daily volume traded is far less than that of established market of the CME.

Currency futures traded in these markets are linked by a mutual offset system. This system allows the traders to open German mark futures contract and/or Japanese yen contract at one exchange and close in the other. For example, trader A, who sells Japanese yen contract to trader B at the IMM can transfer that position and liquidate it at the SIMEX.

The data used in this study consist of daily settlement foreign currency futures prices for the German mark (DM) and the Japanese yen (JY) from November 1, 1993 to June 24, 1996. The total number of observations used in the analysis is 690 for both currency futures. The data set was constructed from quotations on the International Monetary Market (IMM) of the Chicago Mercantile Exchange (CME) and Singapore Monetary Exchange (SIMEX) obtained via Datastream. Prices from the nearby futures contract until the day prior to its last trading day have been used, *i.e.*, maturity date at which the data is rolled over to the next futures contract. The reasons for choosing German mark

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futures contracts and Japanese yen futures contracts in our investigation is because both are traded in the IMM and the SIMEX. Moreover, the trading volumes of these currency futures are large and surpass the other currency futures traded in the IMM, Chicago.

The settlement price on day t { P_t } are transformed to daily returns using the formula R_t = In (P_t / P_{t-1})*100 which produces total observations of 689 for all currency futures. The use of natural logarithmic price changes prevents the nonstationarity in mean and variance of price level in the data series from affecting futures price variablity. As noted by Cavanaugh (1983), the choice between using raw price and the natural logarithm of future prices is largely a matter of econometric convenience. Since sample logarithm of first difference of futures prices or price change or returns often appear to have a more well-behaved distribution than the first difference of the raw series, it will be more convenient to base hypothesis testing on the first difference of natural logarithm of prices. Furthermore, whether futures prices are quoted in terms of U.S. dollars per unit of foreign currency of foreign currency units per U.S. dollar will not significantly affect the analysis.

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As for the timing of trading, the CME and the SIMEX do not operate at the same time since they are in different time zones. Trading at the CME begins about 0720 hours Central Standard Time (CST) and close at about 1400 hours CST for both German mark futures contract and Japanese yen futures contracts. Trading at the SIMEX starts only at about 1900 hours CST and continues till about 0100 hour CST for similar currency futures contracts. There is no trading activity between the CME close and the SIMEX open.

GARCH Modelling of Volatility

In this section, we investigate on the information flow for the CME and SIMEX by estimating whether currency futures contract traded in one market may have an impact on the return as well as volatility of the same currency futures contract traded in other market. The model estimated is an AR(1)-GARCH (1,1) model. Using the approximate maximum likelihood algorithm of Berndt, Hall and Hall and Hausman (1974) of the following model,

$$R_t = \phi_0 + \phi_1 R_{t-1} + \varepsilon_t \qquad \textbf{(1)}$$

$$\varepsilon_t \,|\, \psi_{t-1} \sim N(0, h_t) \qquad \textbf{(2)}$$

$$h_t = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 h_{t-1,} + \delta_1 R^2 \qquad \textbf{(3)}$$

where $R_{t,}$ the natural logarithm return at time t, ψ_{t-1} is the information set at t-1, ε_1 is a stochastic error condition on ψ_{t-1} which is assumed to be normally distributed with zero

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mean and conditional (time varying) variance h_t. For this reason, GARCH models the conditional variance of the error term as a linear function of the lagged squared residuals and the lagged residual conditional variance. The advantage of a GARCH model is that it captures the tendency in financial data for volatility clustering.

To investigate the spillover of volatility from one market to other market, we include squared return series, R^2, in the conditional variance equation (3). If squared return affects price variability, then we would expect to have a positive and significant parameter estimates, δ_1.

1.3 Empirical Results

To test whether the German mark price changes and Japanese yen price changes behaved as stationary series, the Augmented Dickey-Fuller (ADF) specification is applied to the series. The test results for the ADF is reported in Table 1. The null hypothesis of unit root is clearly rejected at 1% significance level for the German mark contract and the Japanese yen contract.

<INSERT TABLE 1>

Summary statistics for the return series of the two currency futures contracts traded at the CME and the SIMEX are reported in Table 2. This includes the distribution parameter of mean, standard deviation, skewness and kurtosis. None of the means are statistically different from zero. The returns series for German mark and Japanese yen contracts show high skewness and kurtosis indicating that the distributions are leptokurtic in nature with thick tails and sharp peaks at the centre compared to the normal distribution possibly due

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to a time-varying variance. Both skewness and excess kurtosis measures should be equal to zero. This finding is consistent with numerous tests applied to distributions of stock returns and spot rates in foreign currencies (Hsieh,1989) and currency futures (Najang, Rahman and Yung, 1992). Table 2 also reports the autocorrelation in the return series and squared return series using the Ljung-Box *Q*-statistics. The Ljung-Box portmanteau test statistic proposed by McLeod and Li (1983) on squared returns over 6, 12 and 24 lags to test for second-order dependence detected an Autoregressive Conditional Heteroscedasticity (ARCH) effect as shown by the presence of serial correlations in all cases. The leptokurtic nature of the distribution and the presence of ARCH effect in the series therefore support the use of the ARCH specification proposed by Engle (1982) and the extended GARCH model by Bollerslev (1986).

<INSERT TABLE 2>

Table 3 presents the estimated AR(1)-GARCH(1,1) models of currency futures returns contracts of the German mark and Japanese yen traded on the IMM and the SIMEX. The parameters in the conditional variance for all currency futures contracts are statistically significant at usual 5 percent level. The presence of integrated GARCH (IGARCH) are to be found in all currency futures since the sum of ($\alpha_1 + \beta_1$) are fairly close to one except for the Japanese yen traded at the IMM which have 0.6807. The fit of the GARCH(1,1) models is further evaluated by examining the standardized residuals. The Ljung-Box Q-statistics of standardized residual squared due to McLeod-Li (1983) do not indicate any further second-order serial dependence for all the series except for the Japanese yen

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futures contract traded at IMM. This indicates the GARCH(1,1) model does a reasonably good job in capturing the return behavior of the volatility. Furthermore, the model reduced the kurtosis considerable.

The results of the estimation of Equations (1) and (3) are reported in Tables 4, 5, 6 and 7. Table 4 reports the transmission of volatility from the IMM to the SIMEX for German mark futures contracts. The coefficient denoted by δ_1 is statistically significant with a *t*-statistic of 1.9616 at 5 % levels. Similar results of volatility transmission are also found for Japanese yen futures contracts as shown in Table 6. The estimated coefficient, $\delta_{1,}$ is statistically significant with a *t*-statistic of 4.7919 at 1 percent Level. The coefficients of α_1 and β_1 are also statistically different from zero at 1 percent level which indicate the existence of the GARCH effects in the conditional variance equation after taking into account the volatility transmission effects. The implication of this finding is that the squared return in the IMM may also be influential in explaining volatility clustering

The estimated results for the transmission of volatility from the SIMEX to the IMM are shown in Table 5 and table 7 for the Japanese yen futures contracts and the German mark futures contracts, respectively. The coefficient of δ_1 is statistically insignificant with a *t*-statistics of 0.00 which indicates that there is no spillover of volatility from the SIMEX to the IMM. Similar result is also found for the German mark futures contract as shown in Table 7. However, the ARCH effects and GARCH effects are statistically high at 1% level.

<INSERT TABLES 4, 5, 6 and 7>

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It can be suggested from the result that the direction of volatility transmission is from the larger market of IMM to smaller market of SIMEX and not vice versa. The results of the present study are similar to the findings of Abhyankar (1995) on the Eurodollar futures.

Test for Model Specifications

The fit of the GARCH (1,1) model with spillover effect included in equation 3 is further evaluated by investigating it's standardised residuals, $z_{i,t} = \varepsilon_{i,t} / \sigma_{i,t}$. The summary statistics of the standardised residuals assuming normal distributions are reported in the last six row of Tables 4, 5, 6 and 7. The results show that volatility persistence as measured by the sum $\alpha_1 + \beta_1$ is very high for all the currency futures contract traded on the IMM and SIMEX. The Ljung-Box Q-statistics of lag 6 and 12, denoted by LB(6) and LB(12), respectively, are not statistically significant at the usual 5 percent level in all cases. Similarly, the Ljung-Box Q-statistics of standardised residual squared due to McLeod-Li (1983) for lag 6 and 12, denoted by $LB^2(6)$ and $LB^2(12)$, respectively, are reduced considerably which suggest second-order serial dependence is removed from all the series completely except for the Japanese yen traded at the IMM. The model also reduces the leptukurtosis considerably from those of return series for most cases. As a result we argue that our model captures the GARCH dynamics adequately.

4. Summary and Conclusions

This paper examines the nature of volatility transmission for two identical currency futures contracts traded at two different time zones but linked with a mutual offset system. The system allows for twenty-four hours trading. The present study firsts test for the stationarity of the return series. The null hypothesis of unit root is rejected for both

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German mark returns and Japanese yen returns at the usual 5 percent significance level. Since the series report the presence of ARCH effects, we employ GARCH specification, which is found to be more appropriate than the linear stochastic model. Our results provide strong evidence of volatility transmission running from the IMM of the CME to the SIMEX for both German mark contracts and Japanese yen contracts. In contrast, volatility does not spillover from the SIMEX to the IMM. The findings of the present study seem to suggest that larger exchange market of the IMM provide a source of information to other smaller exchange market, such as the SIMEX. As a result, traders can take clue from the activity of larger currency futures markets to position themselves in smaller currency futures markets.

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Table 1 Dickey-Fuller (DF)and Augmented Dickey-Fuller (ADF) Test of unit root on German mark returns and Japanese yen returns of IMM and SIMEX contracts. The null hypothesis of unit roots is clearly rejected for both currency futures. Stationarity of currency futures is required for deriving asymptotically meaningful sampling statistics based on the observations.

	IMM ADF t-ratio		SIMEX ADF t-ratio	
	German mark	Japanese yen	German mark	Japanese yen
DF	-27.857**	-27.056**	-27.725**	-25.787**
ADF	-18.5110**	-18.455**	-18.479**	-18.274**

Note: ** Null hypothesis of a unit root is rejected at 1% significance level.

Table 2 Summary statistics for the German mark futures daily returns and Japanese yen futures daily returns traded at both CME and SIMEX.

	CME		SIMEX	
	German mark	Japanese yen	German mark	Japanese yen
No. of Observations	690	690	690	690
Maximum	3.0610	4.2167	3.2111	3.1816
Minimun	-3.2245	-3.5544	-4.3303	-3.5655
Mean	-0.0079	-0.0294	-0.0137	-0.0059
Standard Dev.	0.6233	0.6951	0.6542	0.6965
Skewness	0.0790	0.4977	-0.4122	-0.3316
Kurtosis	4.7048	5.6524	5.8496	3.7957
LB (6)	4.31	4.18	13.04	6.77
LB (12)	8.40	11.90	16.13	13.06
LB (24)	26.89	33.23	21.90	32.24
LB^2 (6)	29.06**	30.58**	33.07**	52.74**
LB^2(12)	49.26**	51.42**	55.50**	60.02**
LB^2(24)	85.18**	88.27**	71.79**	69.91**

Notes: LB(6), LB(12) and LB(24) refer to the Ljung-Box Portmanteau statistic for returns over 6, 12 and 24 lags, respectively. LB2(6), LB2(12) and LB2(24) refer to the Ljung-Box Portmanteau statistic for squared returns over 6, 12 and 24 lags, respectively. ** indicates significance at 1% level.

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Table 3 AR(1)- GARCH(1,1) Parameter estimates for returns and volatility for the German mark futures contracts and Japanese yen futures contracts traded on the SIMEX, Singapore and on the IMM of CME, Chicago.

$$R_t = \phi_0 + \phi_1 R_{t-1} + \varepsilon_t$$

$$h_t = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 h_{t-1}$$

where R_t represents the return for German mark or Japanese yen at time t traded on the IMM or the SIMEX. ε_t is the error term which is distributed as conditionally normal with time varying variance, h_t.

Parameter	CME		SIMEX	
	German mark	Japanese yen	German mark	Japanese yen
ϕ_0	-0.0227 (-1.0146)	-0.0245 (-1.9231)	0.0343 (0.1450)	0.0113 (0.0427)
ϕ_1	-0.0277 (-0.7622)	-0.0505 (-1.1725)	-0.0626 (-1.4952	-0.0040 (-0.0966)
α_0	0.0401 (3.5223)*	0.1550 (5.3551)*	0.0725 (3.3532)*	0.0471 (3.9722)*
α_1	0.0307 (5.5377)*	0.2006 (5.9745)*	0.0383 (4.4613)*	0.0932 (4.3207)*
β_1	0.9581 (129.84)*	0.4801 (5.7437)*	0.9454 (84.485)*	0.8101 (19.712)*
Skewness	0.2472	0.1447	-0.7649	-0.6813
Kurtosis	3.3757	4.3413	6.4633	4.5577
LB (6)	1.7846	2.4461	9.1422	4.4103
LB (12)	5.9987	8.1234	10.4543	10.6296
$LB^2(6)$	4.6084	1.4134	1.0310	1.2696
$LB^2(12)$	5.8698	25.107*	7.9714	1.9998

Notes: Figures in parentheses indicate the *t*-ratio for the parameter. LB(6) and LB(12) refer to the Ljung-Box Portmanteau statistic for returns over 6 and 12 lags, respectively. LB2(6) and LB2(12) refer to the Ljung-Box Portmanteau statistic for squared returns over 6 and 12 lags, respectively. * indicates significance at 5% level.

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Table 4 AR(1)- GARCH(1,1) Parameter estimates for returns and volatility of the German mark futures contract traded on the SIMEX, Singapore with the spillover of squared return from the German mark futures contract traded on the IMM of the CME:

$$R_t = \phi_0 + \phi_1 R_{t-1} + \varepsilon_t$$

$$h_t = \alpha_0 + \alpha_1 \varepsilon^2_{t-1} + \beta_1 h_{t-1,} + \delta_1 R^2_{\text{IMM}}$$

where R_t represents the return for German mark at time t in the SIMEX and R^2_{IMM} is the squared returns for German mark on the IMM.

Parameter		*t*-statistics
ϕ_0	0.6004-02	0.2558
ϕ_1	-0.0600	-1.4304
α_0	0.3094-02	1.3518
α_1	0.0317	4.5244[a]
β_1	0.9559	100.48[a]
δ_1	0.6190	1.9616[b]
Diagnostic Check on the Standardized Residuals		
Skewness	-0.7514	
Kurtosis	6.6034	
LB (6)	9.1587	
LB (12)	10.540	
$LB^2(6)$	1.6361	
$LB^2(12)$	8.3187	

The [a] and [b] indicate significance at 1% and 5%, respectively.

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Table 5 AR(1)- GARCH(1,1) Parameter estimates for returns and volatility of the German mark futures contract traded on the IMM of CME, Chicago with the spillover of squared return from the German mark futures contract traded on the SIMEX, Singapore:

$$R_t = \phi_0 + \phi_1 R_{t-1} + \varepsilon_t$$

$$h_t = \alpha_0 + \alpha_1 \varepsilon^2_{t-1} + \beta_1 h_{t-1,} + \delta_1 R^2_{\text{SIMEX}}$$

where R_t represents the return for German mark at time t on the IMM and R^2_{SIMEX} is the squared returns for German mark on the SIMEX.

Parameter		*t*-statistics
ϕ_0	-0.02278	-1.0139
ϕ_1	-0.0277	-0.7613
α_0	0.4021-02	3.5225[a]
α_1	0.0307	5.5334[a]
β_1	0.9581	129.68[a]
δ_1	0.0	0.0
Diagnostic Check on the Standardized Residuals		
Skewness	0.2470	
Kurtosis	3.7564	
LB (6)	1.7851	
LB (12)	5.9992	
$LB^2(6)$	4.6105	
$LB^2(12)$	5.8721	

Note: The [a] nd [b] indicates significance at 1% and 5%, respectively.

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Table 6 AR(1)- GARCH(1,1) Parameter estimates for returns and volatility of the Japanese yen futures contract traded on the SIMEX, Singapore with the spillover of squared returns from the Japanese yen futures contract traded in IMM of the CME:

$$R_t = \phi_0 + \phi_1 R_{t-1} + \varepsilon_t$$

$$h_t = \alpha_0 + \alpha_1 \varepsilon^2_{t-1} + \beta_1 h_{t-1,} + \delta_1 R^2_{\text{IMM}}$$

where R_t represents the return for Japanese yen at time t on the SIMEX and R^2_{IMM} is the squared returns of Japanese yen on the IMM.

Parameter		*t*-statistics
ϕ_0	0.8560-02	0.3295
ϕ_1	0.2281-02	0.0515
α_0	0.0579	3.4141[a]
α_1	0.1389	4.6683[a]
β_1	0.7185	13.0848[a]
δ_1	0.0268	4.7919[a]
Diagnostic Check on the Standardized Residuals		
Skewness	-0.6197	
Kurtosis	3.9312	
LB (6)	4.0606	
LB (12)	10.268	
$LB^2(6)$	1.2887	
$LB^2(12)$	2.2394	

Note: The [a] and [b] indicates significance at 1% and 5%, respectively.

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Table 7: AR(1)- GARCH(1,1) parameter estimates for returns and volatility of the Japanese yen futures contract traded on the IMM of CME, Chicago with the spillover of squared returns from the Japanese yen futures contract traded on the SIMEX, Singapore:

$$R_t = \phi_0 + \phi_1 R_{t-1} + \varepsilon_t$$

$$h_t = \alpha_0 + \alpha_1 \varepsilon^2_{t-1} + \beta_1 h_{t-1,} + \delta_1 R^2{}_{\text{SIMEX}}$$

where R_t represents the return for Japanese yen at time t on the IMM and $R^2{}_{\text{SIMEX}}$ is the squared returns for Japanese yen on the SIMEX.

Parameter		*t*-statistics
ϕ_0	-0.0337	-1.4719
ϕ_1	-0.0395	-1.0209
α_0	0.4247-02	2.8799[a]
α_1	0.0436	7.3317[a]
β_1	0.9477	142.608[a]
δ_1	0.0	0.0
Diagnostic Check on the Standardized Residuals		
Skewness	0.4074	
Kurtosis	2.6212	
LB (6)	2.3283	
LB (12)	9.2224	
$LB^2(6)$	4.0989	
$LB^2(12)$	7.4526	

Note: The [a] and [b] indicates significance at 1% and 5%, respectively.

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